

Practical 6

Aim: Create Numpy arrays, perform basic operations (addition, multiplication), and apply statistical functions (mean, median, standard deviation).

Source-Code:

```
import numpy as np
arr1 = np.array([10, 20, 30, 40, 50])
arr2 = np.array([1, 2, 3, 4, 5])
print("First Array:")
print(arr1)
print("\nSecond Array:")
print(arr2)
addition = arr1 + arr2
print("\nAddition of Arrays:")
print(addition)
multiplication = arr1 * arr2
print("\nMultiplication of Arrays:")
print(multiplication)
mean_value = np.mean(arr1)
print("\nMean:")
print(mean_value)
median_value = np.median(arr1)
print("\nMedian:")
print(median_value)
std_value = np.std(arr1)
print("\nStandard Deviation:")
print(std_value)
```

Output:

```
First Array:  
[10 20 30 40 50]
```

```
Second Array:  
[1 2 3 4 5]
```

```
Addition of Arrays:  
[11 22 33 44 55]
```

```
Multiplication of Arrays:  
[ 10  40  90 160 250]
```

```
Mean:  
30.0
```

```
Median:  
30.0
```

```
Standard Deviation:  
14.142135623730951
```